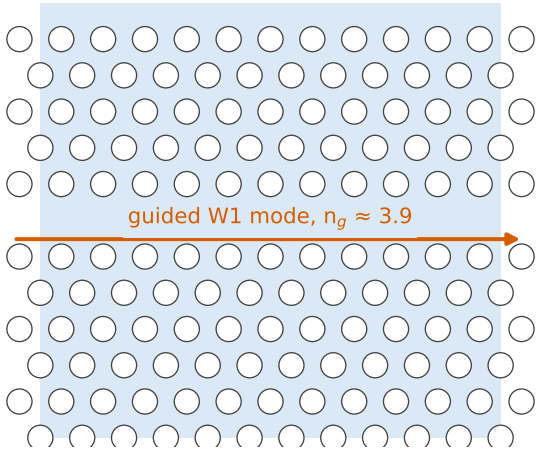
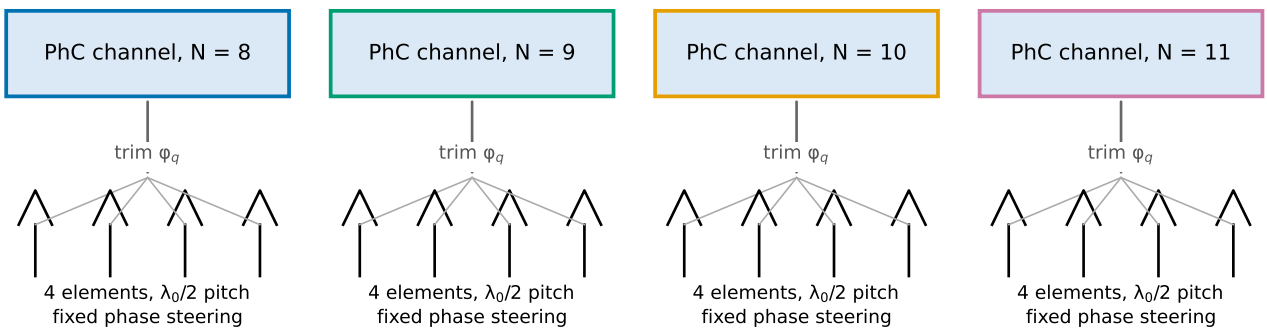
(a) W1 channel: triangular air holes, ε = 12, r/a = 0.30, TE.
2-D FDTD domain (Phases 2–4), N = 8–11 periods; drawn for N = 11



(b) Beamforming model (Phase 6): array factor from the simulated transfer functions $H_m(f)$ (analytic, not FDTD)

subarray pitch $2\lambda_0$; PhC one-period step = 0.981 cycle; required step = 1.000 cycle ($\theta_0 = 30°$)



Not modeled: loss, coupling, bends, disorder, thermal effects, 3-D